

Newman-Penrose Formalism Calculations

Stephani et al., Equation (18.64)

Metric

$$ds^2 = \left(-\frac{h(y)k(y)}{x^2 + 2xy + y^2} \right) dt \otimes dt + \left(\frac{f(x)}{x^2 + 2xy + y^2} \right) d\phi \otimes d\phi + \left(\frac{1}{x^2 f(x) + 2xyf(x) + y^2 f(x)} \right) dx \otimes dx + \frac{1}{(x^2 + 2xy + y^2)h(y)} dy \otimes dy$$

Coordinates

$$(x^\mu) = (t, \phi, x, y)$$

Metric Functions

$$k = A^2 \int \frac{1}{h(y)^{\frac{3}{2}}} dy^2 + 2AB \int \frac{1}{h(y)^{\frac{3}{2}}} dy + B^2$$

Null Tetrad

Covariant components

$$l = \left(\frac{\sqrt{2}\sqrt{h(y)}\sqrt{k(y)}}{2(x+y)} \right) dt + \left(\frac{\sqrt{2}}{2(x+y)\sqrt{h(y)}} \right) dy$$

$$n = \left(\frac{\sqrt{2}\sqrt{h(y)}\sqrt{k(y)}}{2(x+y)} \right) dt + \left(-\frac{\sqrt{2}}{2(x+y)\sqrt{h(y)}} \right) dy$$

$$m = \left(\frac{\sqrt{2}\sqrt{f(x)}}{2(x+y)} \right) d\phi + \left(\frac{i\sqrt{2}}{2(x+y)\sqrt{f(x)}} \right) dx$$

$$\bar{m} = \left(\frac{\sqrt{2}\sqrt{f(x)}}{2(x+y)} \right) d\phi + \left(-\frac{i\sqrt{2}}{2(x+y)\sqrt{f(x)}} \right) dx$$

Contravariant components

$$l = \left(-\frac{\sqrt{2}x}{2\sqrt{h(y)}\sqrt{k(y)}} - \frac{\sqrt{2}y}{2\sqrt{h(y)}\sqrt{k(y)}} \right) dt + \left(\frac{1}{2}\sqrt{2}x\sqrt{h(y)} + \frac{1}{2}\sqrt{2}y\sqrt{h(y)} \right) dy$$

$$n = \left(-\frac{\sqrt{2}x}{2\sqrt{h(y)}\sqrt{k(y)}} - \frac{\sqrt{2}y}{2\sqrt{h(y)}\sqrt{k(y)}} \right) dt + \left(-\frac{1}{2}\sqrt{2}x\sqrt{h(y)} - \frac{1}{2}\sqrt{2}y\sqrt{h(y)} \right) dy$$

$$m = \left(\frac{\sqrt{2}x}{2\sqrt{f(x)}} + \frac{\sqrt{2}y}{2\sqrt{f(x)}} \right) d\phi + \left(\frac{1}{2}i\sqrt{2}x\sqrt{f(x)} + \frac{1}{2}i\sqrt{2}y\sqrt{f(x)} \right) dx$$

$$\bar{m} = \left(\frac{\sqrt{2}x}{2\sqrt{f(x)}} + \frac{\sqrt{2}y}{2\sqrt{f(x)}} \right) d\phi + \left(-\frac{1}{2}i\sqrt{2}x\sqrt{f(x)} - \frac{1}{2}i\sqrt{2}y\sqrt{f(x)} \right) dx$$

Directional Derivatives

$$D = -\frac{\sqrt{2}x}{2\sqrt{h(y)}\sqrt{k(y)}} - \frac{\sqrt{2}y}{2\sqrt{h(y)}\sqrt{k(y)}} \partial_t + \frac{1}{2}\sqrt{2}x\sqrt{h(y)} + \frac{1}{2}\sqrt{2}y\sqrt{h(y)} \partial_y$$

$$\Delta = -\frac{\sqrt{2}x}{2\sqrt{h(y)}\sqrt{k(y)}} - \frac{\sqrt{2}y}{2\sqrt{h(y)}\sqrt{k(y)}} \partial_t - \frac{1}{2}\sqrt{2}x\sqrt{h(y)} - \frac{1}{2}\sqrt{2}y\sqrt{h(y)} \partial_y$$

$$\delta = \frac{\sqrt{2}x}{2\sqrt{f(x)}} + \frac{\sqrt{2}y}{2\sqrt{f(x)}} \partial_\phi + \frac{1}{2}i\sqrt{2}x\sqrt{f(x)} + \frac{1}{2}i\sqrt{2}y\sqrt{f(x)} \partial_x$$

$$\bar{\delta} = \frac{\sqrt{2}x}{2\sqrt{f(x)}} + \frac{\sqrt{2}y}{2\sqrt{f(x)}} \partial_\phi - \frac{1}{2}i\sqrt{2}x\sqrt{f(x)} - \frac{1}{2}i\sqrt{2}y\sqrt{f(x)} \partial_x$$

Spin Coefficients

$$\rho = \frac{1}{2} \sqrt{2} \sqrt{h(y)}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = \frac{1}{2} \sqrt{2} \sqrt{h(y)}$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = \frac{i \sqrt{2} x \frac{\partial}{\partial x} f(x)}{8 \sqrt{f(x)}} + \frac{i \sqrt{2} y \frac{\partial}{\partial x} f(x)}{8 \sqrt{f(x)}} - \frac{1}{4} i \sqrt{2} \sqrt{f(x)}$$

$$\beta = \frac{i \sqrt{2} x \frac{\partial}{\partial x} f(x)}{8 \sqrt{f(x)}} + \frac{i \sqrt{2} y \frac{\partial}{\partial x} f(x)}{8 \sqrt{f(x)}} - \frac{1}{4} i \sqrt{2} \sqrt{f(x)}$$

$$\tau = \frac{1}{2} i \sqrt{2} \sqrt{f(x)}$$

$$\pi = \frac{1}{2} i \sqrt{2} \sqrt{f(x)}$$

$$\epsilon = \frac{\sqrt{2} x \frac{\partial}{\partial y} h(y)}{8 \sqrt{h(y)}} + \frac{\sqrt{2} y \frac{\partial}{\partial y} h(y)}{8 \sqrt{h(y)}} + \frac{\sqrt{2} x \sqrt{h(y)} \frac{\partial}{\partial y} k(y)}{8 k(y)} + \frac{\sqrt{2} y \sqrt{h(y)} \frac{\partial}{\partial y} k(y)}{8 k(y)} - \frac{1}{4} \sqrt{2} \sqrt{h(y)}$$

$$\gamma = \frac{\sqrt{2}x \frac{\partial}{\partial y} h(y)}{8\sqrt{h(y)}} + \frac{\sqrt{2}y \frac{\partial}{\partial y} h(y)}{8\sqrt{h(y)}} + \frac{\sqrt{2}x\sqrt{h(y)} \frac{\partial}{\partial y} k(y)}{8k(y)} + \frac{\sqrt{2}y\sqrt{h(y)} \frac{\partial}{\partial y} k(y)}{8k(y)} - \frac{1}{4}\sqrt{2}\sqrt{h(y)}$$

Ricci Tensor Components

$$\Phi_{00} = -\frac{xh(y) \frac{\partial}{\partial y} k(y)}{4k(y)} - \frac{yh(y) \frac{\partial}{\partial y} k(y)}{4k(y)}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\begin{aligned} \Phi_{11} = & -\frac{1}{8}x^2 \frac{\partial^2}{(\partial x)^2} f(x) - \frac{1}{4}xy \frac{\partial^2}{(\partial x)^2} f(x) - \frac{1}{8}y^2 \frac{\partial^2}{(\partial x)^2} f(x) + \frac{1}{8}x^2 \frac{\partial^2}{(\partial y)^2} h(y) + \frac{1}{4}xy \frac{\partial^2}{(\partial y)^2} h(y) + \frac{1}{8}y^2 \frac{\partial^2}{(\partial y)^2} h(y) + \frac{3x^2 \frac{\partial}{\partial y} h(y) \frac{\partial}{\partial y} k(y)}{16k(y)} \\ & + \frac{3xy \frac{\partial}{\partial y} h(y) \frac{\partial}{\partial y} k(y)}{8k(y)} + \frac{3y^2 \frac{\partial}{\partial y} h(y) \frac{\partial}{\partial y} k(y)}{16k(y)} - \frac{x^2 h(y) \frac{\partial}{\partial y} k(y)^2}{16k(y)^2} - \frac{xy h(y) \frac{\partial}{\partial y} k(y)^2}{8k(y)^2} - \frac{y^2 h(y) \frac{\partial}{\partial y} k(y)^2}{16k(y)^2} + \frac{x^2 h(y) \frac{\partial^2}{(\partial y)^2} k(y)}{8k(y)} \\ & + \frac{xy h(y) \frac{\partial^2}{(\partial y)^2} k(y)}{4k(y)} + \frac{y^2 h(y) \frac{\partial^2}{(\partial y)^2} k(y)}{8k(y)} + \frac{1}{4}x \frac{\partial}{\partial x} f(x) + \frac{1}{4}y \frac{\partial}{\partial x} f(x) - \frac{1}{4}x \frac{\partial}{\partial y} h(y) - \frac{1}{4}y \frac{\partial}{\partial y} h(y) - \frac{xh(y) \frac{\partial}{\partial y} k(y)}{8k(y)} - \frac{yh(y) \frac{\partial}{\partial y} k(y)}{8k(y)} \end{aligned}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = -\frac{xh(y) \frac{\partial}{\partial y} k(y)}{4k(y)} - \frac{yh(y) \frac{\partial}{\partial y} k(y)}{4k(y)}$$

$$\begin{aligned}\Lambda = & -\frac{1}{24}x^2\frac{\partial^2}{(\partial x)^2}f(x) - \frac{1}{12}xy\frac{\partial^2}{(\partial x)^2}f(x) - \frac{1}{24}y^2\frac{\partial^2}{(\partial x)^2}f(x) - \frac{1}{24}x^2\frac{\partial^2}{(\partial y)^2}h(y) - \frac{1}{12}xy\frac{\partial^2}{(\partial y)^2}h(y) - \frac{1}{24}y^2\frac{\partial^2}{(\partial y)^2}h(y) - \frac{x^2\frac{\partial}{\partial y}h(y)\frac{\partial}{\partial y}k(y)}{16k(y)} \\ & - \frac{xy\frac{\partial}{\partial y}h(y)\frac{\partial}{\partial y}k(y)}{8k(y)} - \frac{y^2\frac{\partial}{\partial y}h(y)\frac{\partial}{\partial y}k(y)}{16k(y)} + \frac{x^2h(y)\frac{\partial}{\partial y}k(y)^2}{48k(y)^2} + \frac{xyh(y)\frac{\partial}{\partial y}k(y)^2}{24k(y)^2} + \frac{y^2h(y)\frac{\partial}{\partial y}k(y)^2}{48k(y)^2} - \frac{x^2h(y)\frac{\partial^2}{(\partial y)^2}k(y)}{24k(y)} - \frac{xyh(y)\frac{\partial^2}{(\partial y)^2}k(y)}{12k(y)} \\ & - \frac{y^2h(y)\frac{\partial^2}{(\partial y)^2}k(y)}{24k(y)} + \frac{1}{4}x\frac{\partial}{\partial x}f(x) + \frac{1}{4}y\frac{\partial}{\partial x}f(x) + \frac{1}{4}x\frac{\partial}{\partial y}h(y) + \frac{1}{4}y\frac{\partial}{\partial y}h(y) + \frac{xh(y)\frac{\partial}{\partial y}k(y)}{8k(y)} + \frac{yh(y)\frac{\partial}{\partial y}k(y)}{8k(y)} - \frac{1}{2}f(x) - \frac{1}{2}h(y)\end{aligned}$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\begin{aligned}\Psi_2 = & \frac{1}{12}x^2\frac{\partial^2}{(\partial x)^2}f(x) + \frac{1}{6}xy\frac{\partial^2}{(\partial x)^2}f(x) + \frac{1}{12}y^2\frac{\partial^2}{(\partial x)^2}f(x) + \frac{1}{12}x^2\frac{\partial^2}{(\partial y)^2}h(y) + \frac{1}{6}xy\frac{\partial^2}{(\partial y)^2}h(y) + \frac{1}{12}y^2\frac{\partial^2}{(\partial y)^2}h(y) + \frac{x^2\frac{\partial}{\partial y}h(y)\frac{\partial}{\partial y}k(y)}{8k(y)} + \frac{xy\frac{\partial}{\partial y}h(y)\frac{\partial}{\partial y}k(y)}{4k(y)} \\ & + \frac{y^2\frac{\partial}{\partial y}h(y)\frac{\partial}{\partial y}k(y)}{8k(y)} - \frac{x^2h(y)\frac{\partial}{\partial y}k(y)^2}{24k(y)^2} - \frac{xyh(y)\frac{\partial}{\partial y}k(y)^2}{12k(y)^2} - \frac{y^2h(y)\frac{\partial}{\partial y}k(y)^2}{24k(y)^2} + \frac{x^2h(y)\frac{\partial^2}{(\partial y)^2}k(y)}{12k(y)} + \frac{xyh(y)\frac{\partial^2}{(\partial y)^2}k(y)}{6k(y)} + \frac{y^2h(y)\frac{\partial^2}{(\partial y)^2}k(y)}{12k(y)}\end{aligned}$$

$$\Psi_3 = 0$$

$$\Psi_4 = 0$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "D"